DECISION – WEEK 02

What we completed:

- Implemented the basic **CPU and memory components**.
- Implemented and tested the selected **8 instructions**.
- Developed the basic **FETCH → DECODE → EXECUTE** instruction cycle.
- Developed the basic **HTML user interface** for the simulator.
- Connected and tested the main simulator components.
- Implemented and tested the **Step** and **Run** functions.
- Tested a complete demo program successfully.
- All team members contributed and committed their work.

Team Work:

- CPU & Instruction Execution – Neil Saldanha
- Memory & Stack – Shahala Shahzeen
- Data Structures & Process Management – Thrupthi Anchan
- OS Scheduling & Context Switching – Aman Assadi

What we tested:

- Tested CPU and memory operations.
- Tested the selected instruction set.
- Tested the FETCH → DECODE → EXECUTE cycle.
- Tested Step and Run execution.
- Tested the basic UI.
- Successfully executed a complete demo program.
- Verified the integration of the implemented components.

Accepted:

- Basic CPU and memory implementation accepted.
- Selected instruction set implementation accepted.
- FETCH → DECODE → EXECUTE cycle accepted.
- Basic UI and Step/Run functionality accepted.
- Complete demo program execution successfully accepted.

Not completed:

- Full implementation of processes and PCB.
- Complete CPU scheduling implementation.
- Full integration of scheduling and context switching with the simulator.
- Final UI improvements and complete system integration.
- Final testing and documentation.

Open Issues:

- Further integration is required between process management, scheduling, and CPU execution.
- Context switching needs to be fully implemented and tested.
- Some UI improvements are still required.
- More testing is needed to ensure all components work correctly together.

Concerns:

- Proper integration of CPU execution with process scheduling needs to be ensured.
- Final testing is required before the complete simulator can be accepted.

Next Plan for Week 3:

- Implement processes and PCB.
- Implement CPU scheduling algorithms.
- Implement context switching.
- Integrate scheduling with CPU execution.
- Improve the UI.
- Perform complete system testing.